PRACTICAL 13 – IMPLEMENTATION OF YOUR OWN PING PROGRAM

AIM

To implement a simple ping program to check host reachability and measure round-trip time.

ALGORITHM

1. Read a host name or IP address from the user.
2. Resolve the host name to an IP address.
3. Create a raw socket using the ICMP protocol.
4. Construct and send an ICMP Echo Request packet.
5. Wait for an Echo Reply and measure the round-trip time.
6. Display the reply time or a timeout message, repeat as required, and close the socket.

PYTHON CODE

```
import socket
import struct
import time

host = input("Enter host name or IP address: ")
ip = socket.gethostbyname(host)

sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_ICMP)
sock.settimeout(2)

for i in range(4):
    packet_id = i
    packet = struct.pack("!BBHHH", 8, 0, 0, packet_id, 1)

    checksum = 0
    for j in range(0, len(packet), 2):
        checksum += (packet[j] << 8) + packet[j + 1]

    checksum = (checksum >> 16) + (checksum & 0xffff)
    checksum = ~checksum & 0xffff

    packet = struct.pack("!BBHHH", 8, 0, checksum, packet_id, 1)
    start = time.time()
    sock.sendto(packet, (ip, 0))

    try:
        sock.recvfrom(1024)
        end = time.time()
        print("Reply from", ip, "time =", round((end - start) * 1000, 2), "ms")
    except socket.timeout:
        print("Request timed out")

sock.close()
```

INPUT

Enter a host name or IP address, for example: 8.8.8.8

OUTPUT

The program displays replies with round-trip time in milliseconds or “Request timed out” if no response is received.

RESULT

A basic ping program was implemented to send ICMP requests and display response times.